StockYard

Inventory and sales, made for makers who outgrew spreadsheets.

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github.com/BDNIKKO/stockyard

THE PROBLEM

Spreadsheets break. Enterprise tools overcharge.

Solo makers selling on Etsy, at markets, and from their own sites hit a wall once inventory grows.

1

Counts drift

Sell a mug at a craft fair and Etsy still shows it in stock. Someone has to fix it by hand.

2

Profit is a guess

Revenue is easy to see. What each channel actually earns after cost of goods is not.

$50

Per month, for features they'll never touch

The usual escape from the spreadsheet is a heavy platform built for businesses ten times the size.

THE SOLUTION

One place for stock, sales, and the numbers

Every capability from the proposal's MVP shipped.

Product catalog

- SKU, photo, cost, price, stock
- Edit any time
- Archive without losing history

Sales ledger

- Log product, channel, quantity, price
- Stock drops automatically
- Filter by date and channel

Your own channels

- Etsy, markets, direct, pop-ups
- Rename or add anytime
- Starter set on sign-up

Dashboard

- Month revenue, units, profit
- Channel breakdown chart
- Low-stock restock list

Private by design

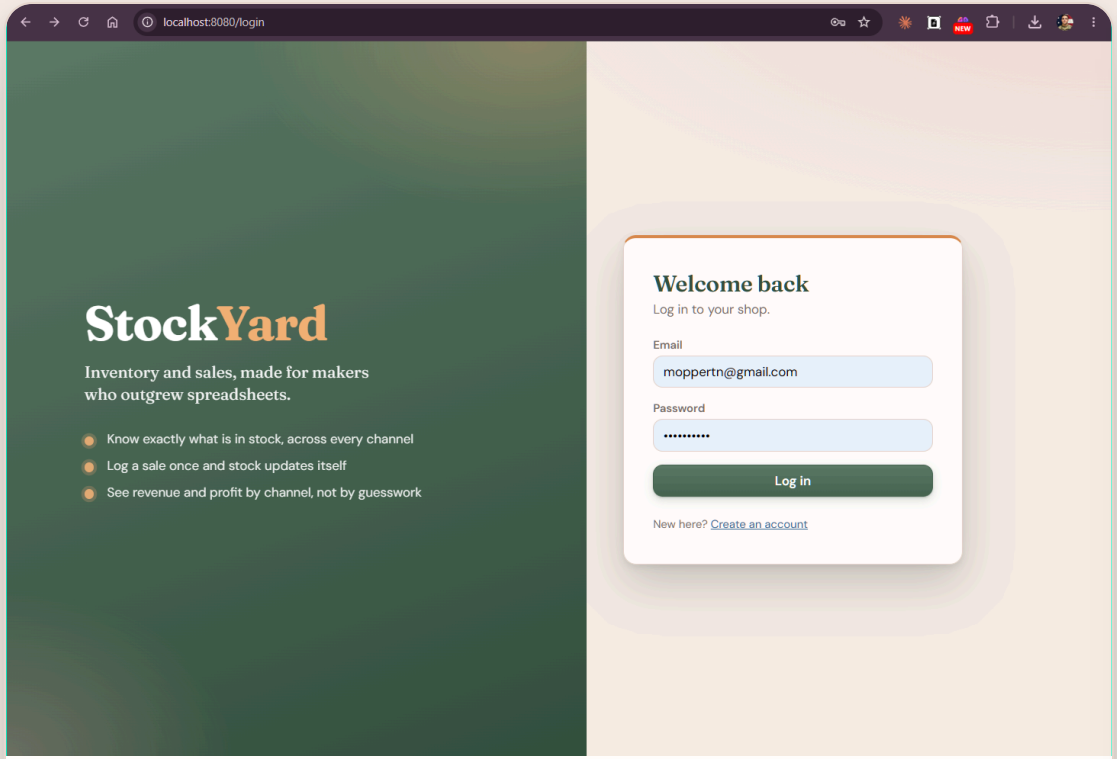
- Secure sign-up and login
- Stays signed in on reload
- Shops can't see each other

Works anywhere

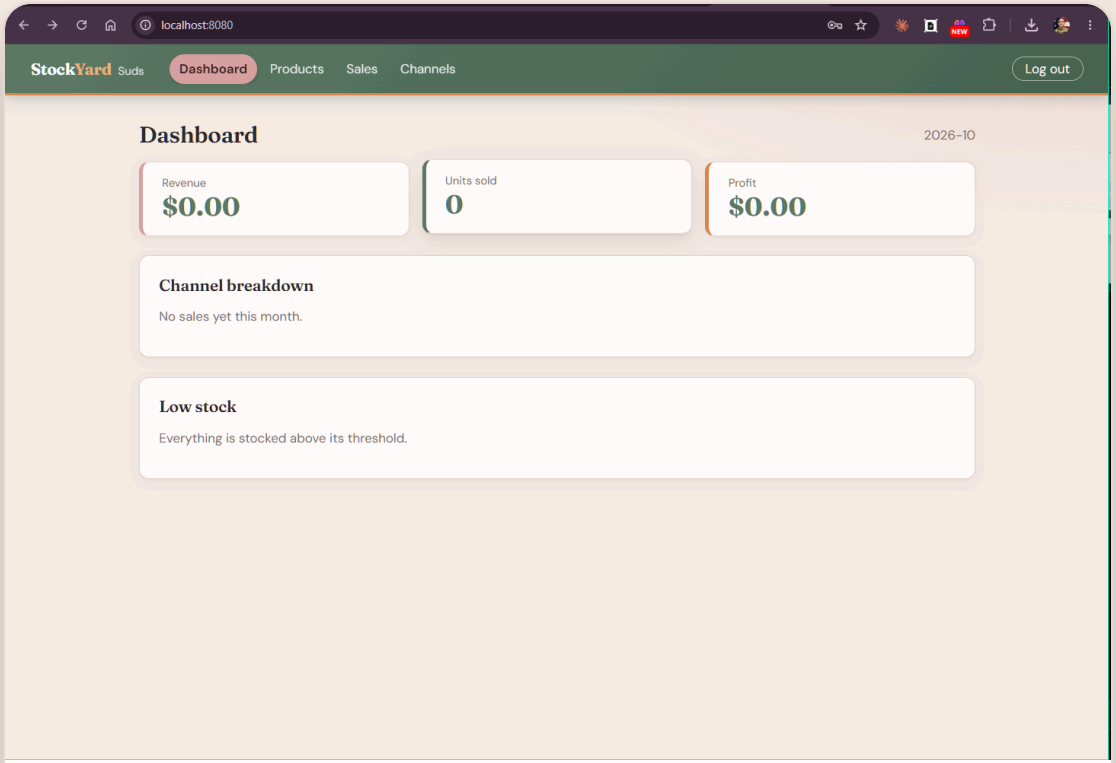
- Phone, tablet, desktop
- Breakpoints at 480, 768, 1024

LIVE DEMO

See it working



Sign in or create a shop

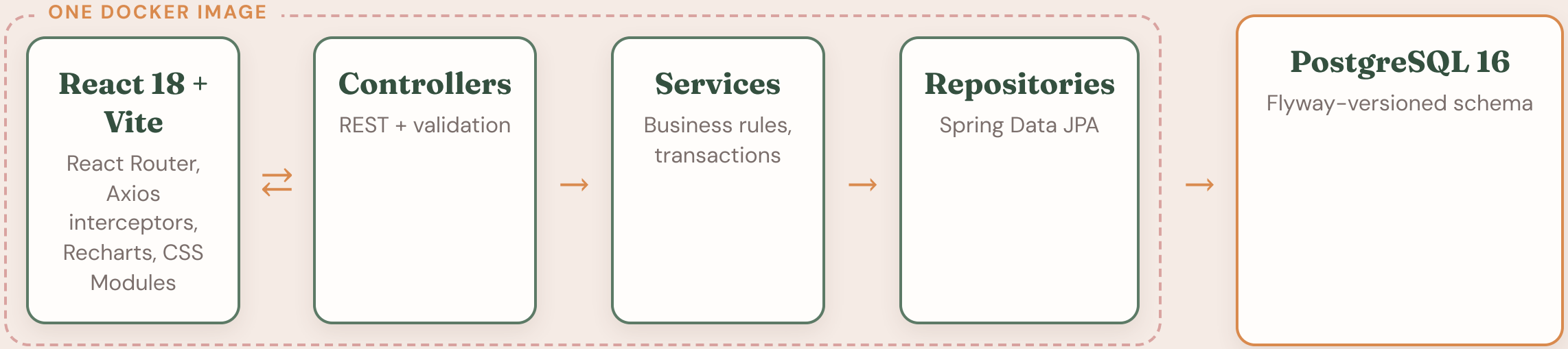


Dashboard: revenue, units, profit, channel chart, low stock

ARCHITECTURE

Layered, simple to run

A React single-page app talks JSON over HTTP to a layered Spring Boot API backed by PostgreSQL.



Java 21 · Spring Boot 3.5 · Spring Security + JWT · DTO records · OpenAPI docs · Docker Compose · GitHub Actions CI

CODE WALKTHROUGH · ONE REQUEST, END TO END

Logging a sale

SALECONTROLLER.JAVA → SALESERVICE.JAVA (TRIMMED)

```
@PostMapping
public SaleResponse record(@AuthenticationPrincipal MakerPrincipal me,
                           @Valid @RequestBody SaleRequest req) {
    return service.record(me.id(), req);
}

@Transactional
public SaleResponse record(Long makerId, SaleRequest req) {
    Product product = products.findByIdAndMakerId(req.productId(), makerId)
        .orElseThrow(() -> ApiException.notFound("Product"));
    // channels are looked up the same way

    if (products.decrementStock(product.getId(), makerId, qty) == 0) {
        throw ApiException.conflict("Not enough stock for " + sku);
    }

    Sale sale = new Sale(maker, product, channel, qty, price);
    // the Sale constructor copies today's unit cost
    return SaleResponse.of(sales.save(sale));
}
```

1 The controller stays thin

Spring Security hands it the signed-in maker, and **@Valid** rejects bad input before any business logic runs. Everything else is delegated.

2 Ownership on every lookup

Products and channels are fetched by id **and** maker. Someone else's product simply doesn't exist for you: a 404, never a data leak.

3 One guarded statement decides

The database lowers stock only if enough remains. Zero rows updated means the sale is refused with a 409. No read-then-write race.

4 Ledger in the same transaction

The sale copies the unit cost at that moment, then saves. **@Transactional** makes stock and ledger commit together, or not at all.

CODE WALKTHROUGH · SECURITY

Stateless JWT authentication

SECURITYCONFIG · JWTAUTHFILTER · JWTSERVICE (TRIMMED)

```
.sessionManagement(s → s.sessionCreationPolicy(STATELESS))
.addFilterBefore(jwtFilter, UsernamePasswordAuthenticationFilter.class)
.authorizeHttpRequests(a → a
    .requestMatchers("/api/auth/register", "/api/auth/login").permitAll()
    .requestMatchers("/api/**").authenticated()
    .anyRequest().permitAll())

// JwtAuthFilter
String header = request.getHeader(HttpHeaders.AUTHORIZATION);
if (header ≠ null && header.startsWith("Bearer ")) {
    jwt.parse(header.substring(7)).ifPresent(principal →
        SecurityContextHolder.getContext().setAuthentication(auth));
}

// JwtService
Jwts.builder().subject(String.valueOf(makerId))
    .claim("email", email)
    .expiration(Date.from(now.plus(ttl)))
    .signWith(key).compact();
```

1 No server sessions

Nothing is remembered on the server between requests. Each request carries its own proof of identity, so the API scales and restarts cleanly.

2 A filter checks every call

It reads the Bearer token, verifies the signature and expiry, and puts the signed-in maker into the security context.

3 Locked by default

Only sign-up and login are open. Every other **/api** route requires a valid token; everything else is just the React app.

4 Signed, expiring tokens

Tokens carry the maker id and an expiry, signed with a secret from the environment. Passwords are stored only as BCrypt hashes.

CODE WALKTHROUGH · DATABASE

A schema that protects itself

V1__INIT.SQL · FLYWAY MIGRATION (EXCERPT)

```
CREATE TABLE sales (
    id BIGINT GENERATED BY DEFAULT AS IDENTITY PRIMARY KEY,
    maker_id   BIGINT NOT NULL REFERENCES makers(id) ON DELETE CASCADE,
    product_id BIGINT NOT NULL REFERENCES products(id),
    channel_id BIGINT NOT NULL REFERENCES channels(id),
    quantity   INTEGER NOT NULL CHECK (quantity > 0),
    unit_price NUMERIC(12,2) NOT NULL CHECK (unit_price ≥ 0),
    unit_cost  NUMERIC(12,2) NOT NULL CHECK (unit_cost ≥ 0),
    sold_at TIMESTAMP WITH TIME ZONE NOT NULL  ...
);

CREATE INDEX idx_sales_maker_date ON sales(maker_id, sold_at);

-- products table
stock INTEGER NOT NULL CHECK (stock ≥ 0),
CONSTRAINT uq_products_maker_sku UNIQUE (maker_id, sku)
```

1 Normalized and linked

Each sale points to its maker, product and channel with foreign keys. No duplicated names, no orphan rows.

2 Rules live in the database

No negative stock or prices, quantity above zero, one SKU per maker. Even a bug in the app can't write bad data.

3 Cost is snapshotted

Each sale stores the cost at the time. Edit a product's cost later and last month's profit stays correct.

4 Indexed for the questions we ask

Maker plus date range is exactly how the dashboard and ledger filter. Flyway applies this same schema in every environment.

CODE WALKTHROUGH · FRONT END

From API to a responsive chart

DASHBOARD.JSX · DASHBOARD.MODULE.CSS · TOKENS.CSS

```
useEffect(() => {
  api.get('/api/dashboard').then((r) => setData(r.data))
     .catch((e) => setError(describeError(e).message));
}, []);

const chartData = data.channels.map((c) => ({
  name: c.channelName, Revenue: Number(c.revenue),
  Profit: Number(c.revenue) - Number(c.cost),
}));

<ResponsiveContainer width="100%" height="100%">
  <BarChart data={chartData}>
    <Bar dataKey="Revenue" fill="#6f9a7c" />
    <Bar dataKey="Profit"  fill="#d98a4e" />

.chart { height: 240px; }
@media (min-width: 480px) { .chart { height: 280px; } }
@media (min-width: 768px) { .chart { height: 320px; } }
/* tokens.css */ --color-sage: #5c7a66; --color-orange: #d98a4e;
```

1 One shared API client

The page just calls the API. An Axios interceptor already attached the token and handles expired sessions in one place.

2 Shape data for the chart

The server sends totals; the page derives profit per channel and hands Recharts a simple list.

3 Charts that resize

The chart fills a container whose height changes by breakpoint, so it never clips on a phone.

4 Tokens, not a UI library

CSS Modules over shared design tokens. The whole green-and-orange re-theme was a handful of variables.

CODE WALKTHROUGH · PROOF

Tests that try to break it

SALESFLOWTEST.JAVA (SIMPLIFIED)

```
// 10 in stock. Sell 8.
send(post("/api/sales"), token, sale(productId, channelId, 8), 201);

// Only 2 left. Selling 3 must fail.
send(post("/api/sales"), token, sale(productId, channelId, 3), 409);

assertThat(product.get("stock").asInt()).isEqualTo(2);
assertThat(send(get("/api/sales"), token, null, 200)).hasSize(1);

assertThat(dash.get("revenue")).isEqualByComparingTo("144.00");
assertThat(dash.get("profit")).isEqualByComparingTo("112.00");

// Another shop tries to sell this product
send(post("/api/sales"), otherMaker, sale(productId, theirChannel, 1), 404);
```

```
Tests run: 3, Failures: 0, Errors: 0  ·  BUILD SUCCESS
```

1 A normal sale works

Ten units in stock, eight sold: the request succeeds with a 201.

2 Overselling is refused

Asking for three when two remain returns a 409 Conflict.

3 And nothing changed

Stock is still two and the ledger still has one entry. A failed sale leaves no trace.

4 The math is right

8 sold at $18 is $144 revenue; with a $4 cost, $112 profit. The dashboard matches to the cent.

5 Shops stay separate

A second maker can't sell the first maker's product: 404. Isolation is tested, not assumed.

QUALITY AND DELIVERY

Containerized, automated, one command to run

Automated checks

- Backend tests with Maven
- Front-end production build
- GitHub Actions on every push

Anyone can run it

- **docker compose up --build**
- Open localhost:8080
- Postgres included

Ready to host

- Single Docker image
- Render blueprint in the repo
- Secrets from environment

REFLECTION

What changed, and what I learned

Proposal	What I shipped	Why
Azure App Service + Static Web Apps	One Docker image, Render blueprint	Anyone can run or view it without cloud credentials
Caching layer for reads	Aggregates computed in PostgreSQL	Fast enough at this scale; less to maintain
Azure Blob image storage	Local disk behind a storage class	Simple now; one class to swap later

Lesson: push consistency into the database, ship the simplest architecture that works, and keep the docs honest about what's real.

StockYard

Next up: restock email alerts, CSV import, a public catalog page, and tags.

Thank you. Questions?

github.com/BDNIKKO/stockyard

Live: stockyard-ucr9.onrender.com

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